

ADAMSON UNIVERSITY
COLLEGE OF SCIENCE
COMPUTER SCIENCE DEPARTMENT
ACADEMIC SERVICE LEARNING

BASKETBALL COURT SCHEDULING SYSTEM

IN COOPERATION WITH
BARANGAY STO. NIÑO, PARAÑAQUE CITY

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The researchers express their sincere gratitude to Adamson University for giving them the opportunity to take part in Academic Service Learning and to apply computer science knowledge in a community-based setting.

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The group acknowledges the guidance of [INSERT PROFESSOR/ASL INSTRUCTOR NAME HERE], whose instruction and feedback supported the completion of the project and this documentation.

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CHAPTER 1: INTRODUCTION

1.1 BACKGROUND OF THE STUDY

The basketball court in Barangay Sto. Niño, Parañaque City serves as a community recreation space where residents, youth groups, and sports teams gather for practice, friendly games, and social interaction. Because the court is a shared barangay facility, its availability affects many residents who depend on it for organized and informal community activities.

The project proposal identifies scheduling conflicts as the main concern. When court use is handled informally, two or more groups may expect to use the same time slot, and barangay personnel may need to manually check whether a requested schedule is available. This can lead to misunderstandings, repeated checking, and difficulty maintaining a clear record of reservations.

To address this need, the Basketball Court Scheduling System was developed as an offline, computer-based scheduling system for the barangay office. Residents do not reserve remotely; they coordinate directly with authorized barangay personnel, who encode reservation details, monitor available and reserved slots, update reservation status, and keep records in the local database.

An organized reservation system supports both barangay personnel and residents. Personnel can check the calendar before approving a request, prevent overlapping reservations, track missed, cancelled, and completed records, and review activity logs. Residents benefit from clearer scheduling and fairer access to the basketball court under barangay policy.

The project is also prepared for ISO/IEC 25010-style evaluation through evidence of functional suitability, reliability, usability, security, maintainability, and portability. The documentation does not assign evaluation scores because no verified score sheet or completed evaluation results were found in the available source files.

1.1.1 THE HISTORY OF BARANGAY STO. NIÑO

[INSERT VERIFIED HISTORY OF BARANGAY STO. NIÑO, PARAÑAQUE CITY HERE]

This section requires verified barangay-provided historical information. No verified history was found in the repository, proposal, slides, or provided ASL reference files.

1.1.2 MISSION, GOAL & VISION

Mission	[INSERT VERIFIED BARANGAY MISSION HERE]
Goal	[INSERT VERIFIED BARANGAY GOAL HERE]
Vision	[INSERT VERIFIED BARANGAY VISION HERE]

1.2 ORGANIZATIONAL STRUCTURE

[INSERT BARANGAY STO. NIÑO ORGANIZATIONAL STRUCTURE HERE]

No official organizational chart or verified list of barangay officials was found in the source files inspected.

1.3 STATEMENT OF THE PROBLEM

The system responds to the need for a clear and centralized method of managing basketball court reservations inside the barangay office.

- Scheduling Conflicts - Informal scheduling can allow multiple groups to expect access to the court at the same time.
- Inefficient Manual Checking - Personnel may need to repeatedly verify whether a date and time are still available.
- Lack of Centralized Reservation Records - Without a system, reservation details, status changes, and requester information can be harder to retrieve and update.
- Difficulty Monitoring Available Time Slots - Barangay personnel need a clear view of reserved, available, missed, cancelled, completed, and blocked periods.
- Limited Access Due to Offline Barangay-Office Setup - The system is intentionally limited to authorized personnel on the installed barangay office computer.

1.4 OBJECTIVES OF THE STUDY

- To provide an organized reservation system for the Barangay Sto. Niño basketball court.
- To prevent scheduling overlaps and double bookings through application validation and database safeguards.
- To maximize facility utilization by giving barangay personnel a clear view of reserved and available court time slots.
- To allow authorized barangay personnel to record, update, and manage reservations efficiently.
- To maintain systematic and updated reservation records in a local MySQL/MariaDB database.
- To support accountability through activity logs for reservation, account, and schedule-related actions where implemented.

1.5 SCOPE AND LIMITATIONS

SCOPE

- Monitoring available basketball court schedules through dashboard and calendar/schedule views.
- Recording reservation requests submitted by residents through authorized barangay personnel.
- Displaying reserved, available, missed, cancelled, completed, maintenance, and public-use blocked slots where applicable.
- Managing reservation records through the administrative staff interface, including reservation details, status changes, filtered lists, print views, and CSV exports.

- Suggesting the nearest available time when the requested schedule is not available, as implemented in the dashboard/API availability logic.
- Marking missed reservations through supported reservation status workflows.
- Managing Admin and Staff accounts, including account creation, activation/deactivation, and password changes.
- Maintaining activity logs for implemented reservation, account, court-policy, and schedule-block actions.
- Providing local backup, restore, readiness, and sign-off tooling through maintenance scripts rather than an in-app backup interface.

LIMITATIONS

- The project is designed for offline deployment based on the barangay preference recorded in the proposal and slides.
- The normal deployment is accessible only on the installed barangay office computer unless the office later configures local-network access.
- Residents cannot reserve remotely; they must coordinate with barangay personnel.
- Scheduling follows Barangay Sto. Niño policies and any policy changes must be confirmed by the barangay.
- Multi-device, public internet, cloud database, online payment, email, and SMS notification features are not included in the current system.
- Actual barangay-office PC setup, physical printer output, signed turnover, and completed evaluation scores remain external items that must be verified on site.

CHAPTER 2: DATABASE & PROGRAM STRUCTURE

2.1 PROGRAM FLOWCHART

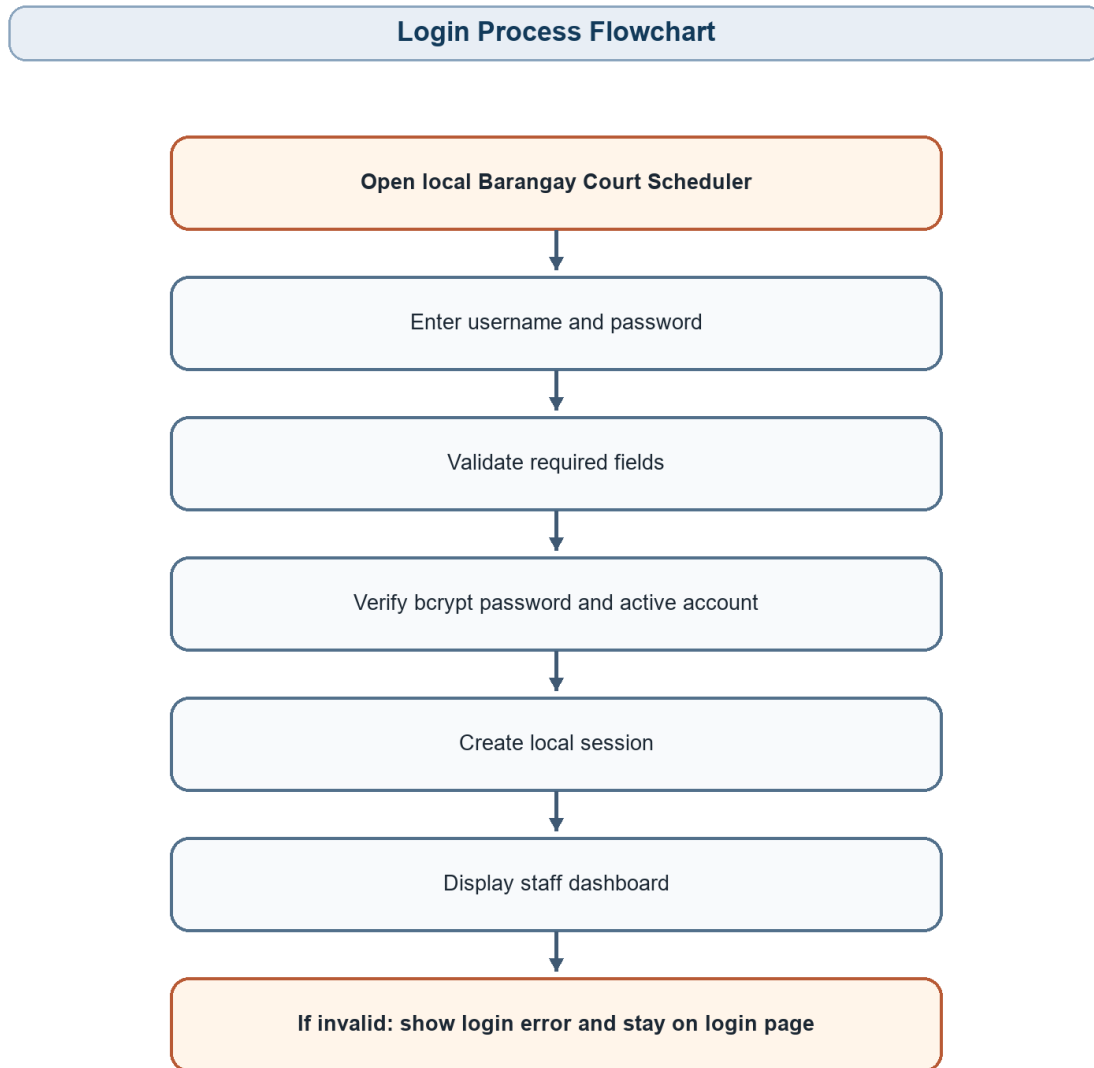


Figure 2.1. Login Process Flowchart

Dashboard and Navigation Flowchart

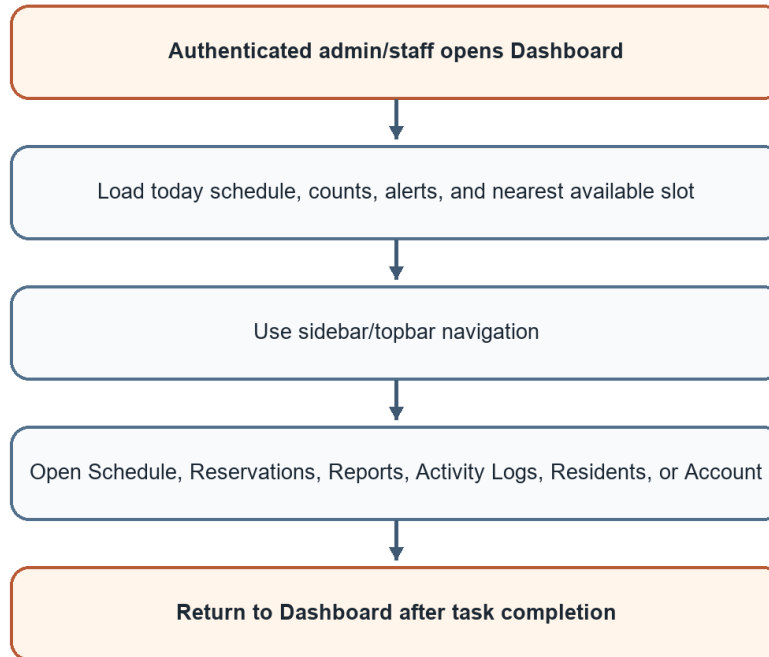


Figure 2.2. Dashboard and Navigation Flowchart

Reservation Creation Flowchart

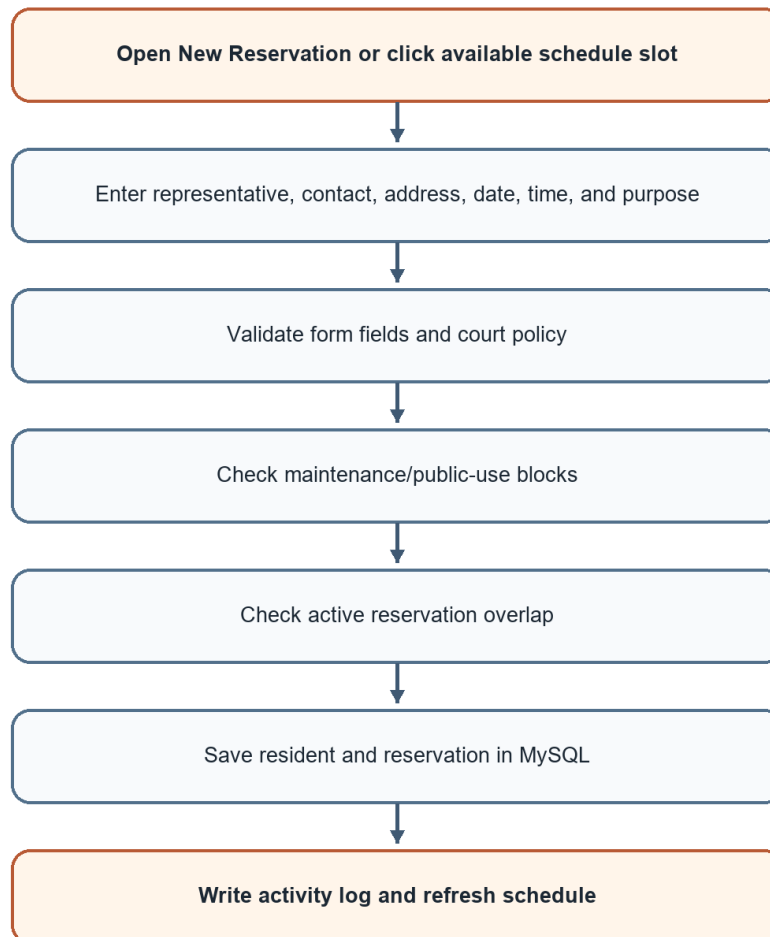


Figure 2.3. Reservation Creation Flowchart

Schedule and Calendar Monitoring Flowchart

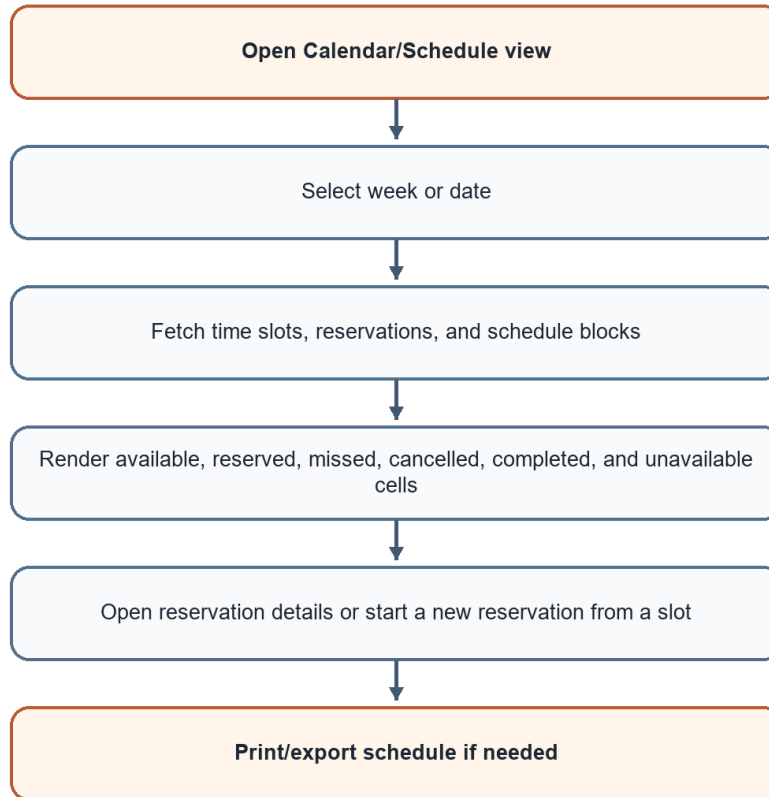


Figure 2.4. Schedule and Calendar Monitoring Flowchart

Account Management Flowchart

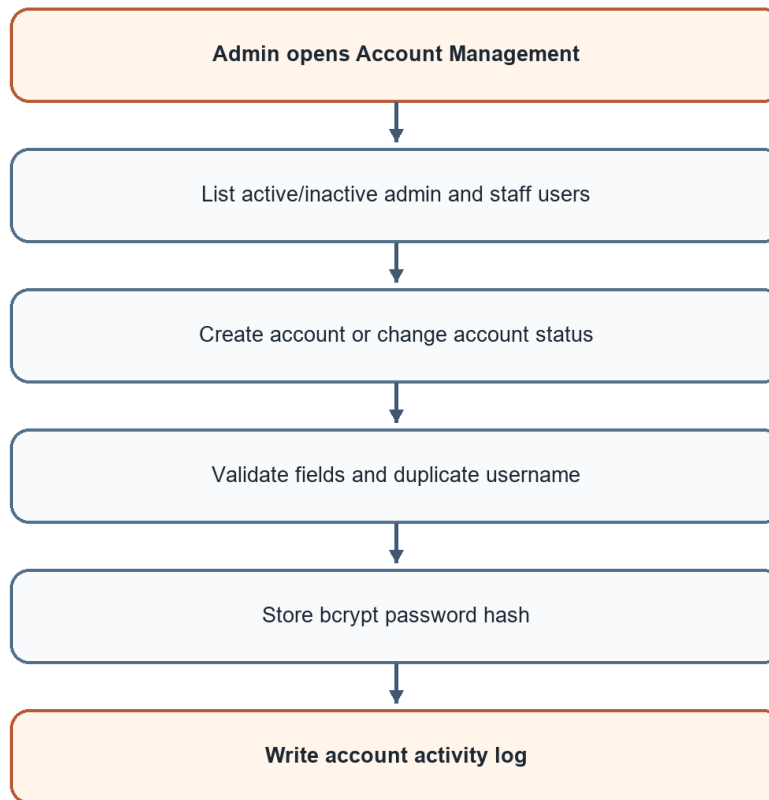


Figure 2.5. Account Management Flowchart

Activity Logs Flowchart

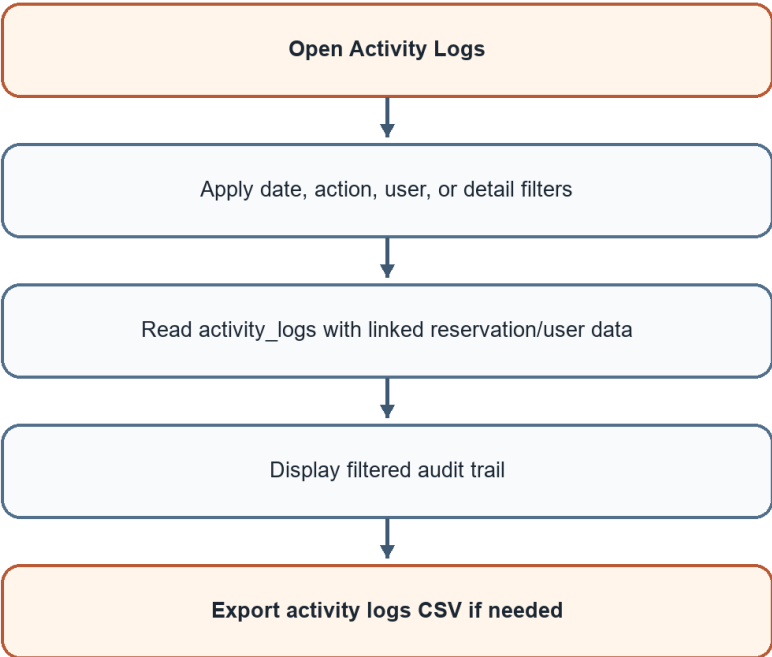


Figure 2.6. Activity Logs Flowchart

Backup, Restore, and Maintenance Tooling Flowchart

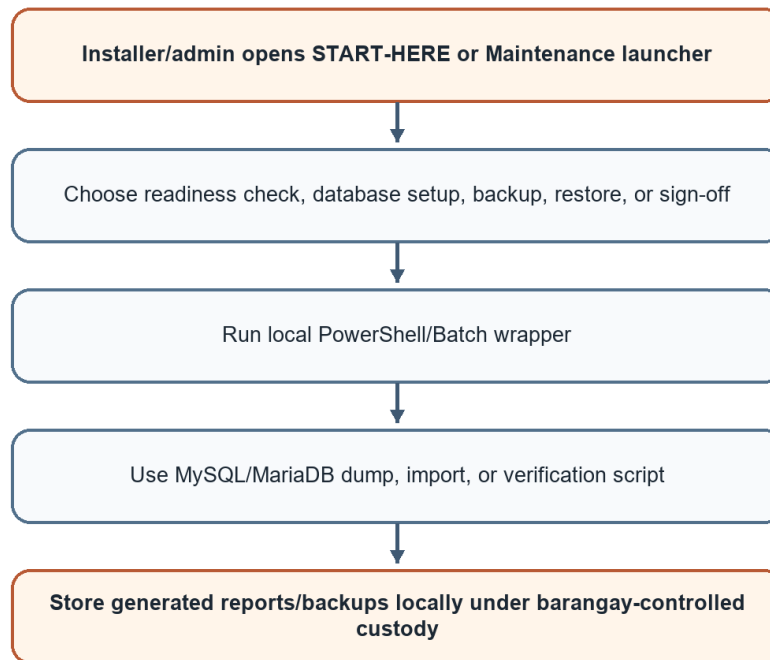


Figure 2.7. Backup, Restore, and Maintenance Tooling Flowchart

2.2 ENTITY RELATIONSHIP NOTATION

The current implementation uses a local MySQL/MariaDB database named `barangay_court_scheduler`. The schema is defined in `database/schema.sql` and uses InnoDB tables, utf8mb4/utf8mb4_unicode_ci encoding, foreign keys, indexes, CHECK constraints, reservation reference sequences, schedule blocks, and overlap-prevention triggers.

users

Field Name	Data Type	Key/Constraint	Description
<code>user_id</code>	BIGINT UNSIGNED	PK, AUTO_INCREMENT	Unique user account identifier.
<code>full_name</code>	VARCHAR(120)	NOT NULL	Name of the authorized barangay user.
<code>contact_no</code>	VARCHAR(30)	NULL	Optional staff contact number.
<code>username</code>	VARCHAR(60)	NOT NULL, UNIQUE	Login username.
<code>password_hash</code>	VARCHAR(255)	NOT NULL	Bcrypt password hash; plaintext passwords are not stored.

role	VARCHAR(20)	CHECK ADMIN/STAFF	Role used for permissions.
account_status	VARCHAR(20)	CHECK ACTIVE/INACTIVE	Controls account use.
created_at	TIMESTAMP	DEFAULT CURRENT_TIMESTAMP	Creation timestamp.
updated_at	TIMESTAMP	ON UPDATE CURRENT_TIMESTAMP	Latest update timestamp.

residents

Field Name	Data Type	Key/Constraint	Description
resident_id	BIGINT UNSIGNED	PK, AUTO_INCREMENT	Unique resident/requester identifier.
full_name	VARCHAR(140)	NOT NULL, INDEX	Resident or representative name.
contact_no	VARCHAR(30)	NOT NULL, INDEX	Resident or representative contact number.
address	VARCHAR(255)	NOT NULL	Resident address.
group_name	VARCHAR(140)	NULL	Optional team or group name.
notes	TEXT	NULL	Optional resident notes.
created_at	TIMESTAMP	DEFAULT CURRENT_TIMESTAMP	Record creation timestamp.
updated_at	TIMESTAMP	ON UPDATE CURRENT_TIMESTAMP	Latest update timestamp.

reservation_statuses

Field Name	Data Type	Key/Constraint	Description
status_id	SMALLINT UNSIGNED	PK	Status identifier.
status_code	VARCHAR(30)	NOT NULL, UNIQUE	Machine-readable status.
status_name	VARCHAR(50)	NOT NULL	Display label.
is_blocking	TINYINT(1)	CHECK 0/1	Whether status blocks new reservations.
display_order	SMALLINT UNSIGNED	NOT NULL	Display order.

time_slots

Field Name	Data Type	Key/Constraint	Description
slot_id	SMALLINT UNSIGNED	PK, AUTO_INCREMENT	Unique slot identifier.
name	VARCHAR(50)	NOT NULL	Display name.
start_time	TIME	NOT NULL, UNIQUE WITH	Slot start time.

		end_time	
end_time	TIME	NOT NULL, CHECK end > start	Slot end time.
display_order	SMALLINT UNSIGNED	NOT NULL	Display order.
is_active	TINYINT(1)	CHECK 0/1	Whether slot is active.

court_settings

Field Name	Data Type	Key/Constraint	Description
setting_key	VARCHAR(80)	PK	Unique setting key.
setting_value	VARCHAR(255)	NOT NULL	Configured value.
description	VARCHAR(255)	NULL	Setting explanation.
updated_at	TIMESTAMP	ON UPDATE CURRENT_TIMESTAMP	Latest update timestamp.

reservation_reference_sequences

Field Name	Data Type	Key/Constraint	Description
reference_year	SMALLINT UNSIGNED	PK	Year for BCS references.
next_sequence	BIGINT UNSIGNED	NOT NULL	Next reservation number for the year.
updated_at	TIMESTAMP	ON UPDATE CURRENT_TIMESTAMP	Latest update timestamp.

reservations

Field Name	Data Type	Key/Constraint	Description
reservation_id	BIGINT UNSIGNED	PK, AUTO_INCREMENT	Unique reservation identifier.
reference_no	VARCHAR(20)	NOT NULL, UNIQUE	BCS-YYYY-000001 reference.
resident_id	BIGINT UNSIGNED	FK residents	Linked resident/requester.
time_slot_id	SMALLINT UNSIGNED	FK time_slots, NULL	Optional display time-slot reference.
status_id	SMALLINT UNSIGNED	FK reservation_statuses	Current reservation status.
approved_by_user_id	BIGINT UNSIGNED	FK users, NULL	Approving user when applicable.
created_by_user_id	BIGINT UNSIGNED	FK users	User who encoded reservation.
reservation_date	DATE	NOT NULL, INDEX	Date reserved.
start_time	TIME	NOT NULL	Reservation start time.
end_time	TIME	NOT NULL, CHECK end > start	Reservation end time.

purpose	VARCHAR(120)	NOT NULL	Purpose of court use.
remarks	TEXT	NULL	Optional remarks.
created_at	TIMESTAMP	DEFAULT CURRENT_TIMESTAMP	Creation timestamp.
updated_at	TIMESTAMP	ON UPDATE CURRENT_TIMESTAMP	Latest update timestamp.

schedule_blocks

Field Name	Data Type	Key/Constraint	Description
block_id	BIGINT UNSIGNED	PK, AUTO_INCREMENT	Unique unavailable-range identifier.
block_category	VARCHAR(30)	CHECK MAINTENANCE/PUBLIC_USE	Type of unavailable range.
block_type	VARCHAR(40)	NOT NULL	Specific reason category.
mode	VARCHAR(30)	CHECK WHOLE_DAY/TIME_RANGE/FROM_TIME_ONWARD	How time range is applied.
reservation_date	DATE	NOT NULL, INDEX	Date affected.
start_time	TIME	NOT NULL	Block start time.
end_time	TIME	NOT NULL, CHECK end > start	Block end time.
reason	VARCHAR(255)	NOT NULL	Reason entered by personnel.
created_by_user_id	BIGINT UNSIGNED	FK users	User who created block.
deactivated_by_user_id	BIGINT UNSIGNED	FK users, NULL	User who deactivated block.
is_active	TINYINT(1)	CHECK 0/1	Whether block is active.
created_at	TIMESTAMP	DEFAULT CURRENT_TIMESTAMP	Creation timestamp.
updated_at	TIMESTAMP	ON UPDATE CURRENT_TIMESTAMP	Latest update timestamp.
deactivated_at	TIMESTAMP	NULL	Deactivation timestamp.

activity_logs

Field Name	Data Type	Key/Constraint	Description
log_id	BIGINT UNSIGNED	PK, AUTO_INCREMENT	Unique activity log identifier.
reservation_id	BIGINT UNSIGNED	FK reservations, NULL	Reservation connected to action.
user_id	BIGINT UNSIGNED	FK users, NULL	User connected to action.
action	VARCHAR(80)	NOT NULL, INDEX	Action label.
details	TEXT	NULL	Additional details.

created_at	TIMESTAMP	DEFAULT CURRENT_TIMESTAMP, INDEX	Time action was logged.
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Database safeguards include MySQL triggers named *prevent_reservation_overlap_before_insert* and *prevent_reservation_overlap_before_update*. These reject overlapping active reservations at the database level when the new or updated status is blocking.

2.3 ENTITY RELATIONSHIP DIAGRAM

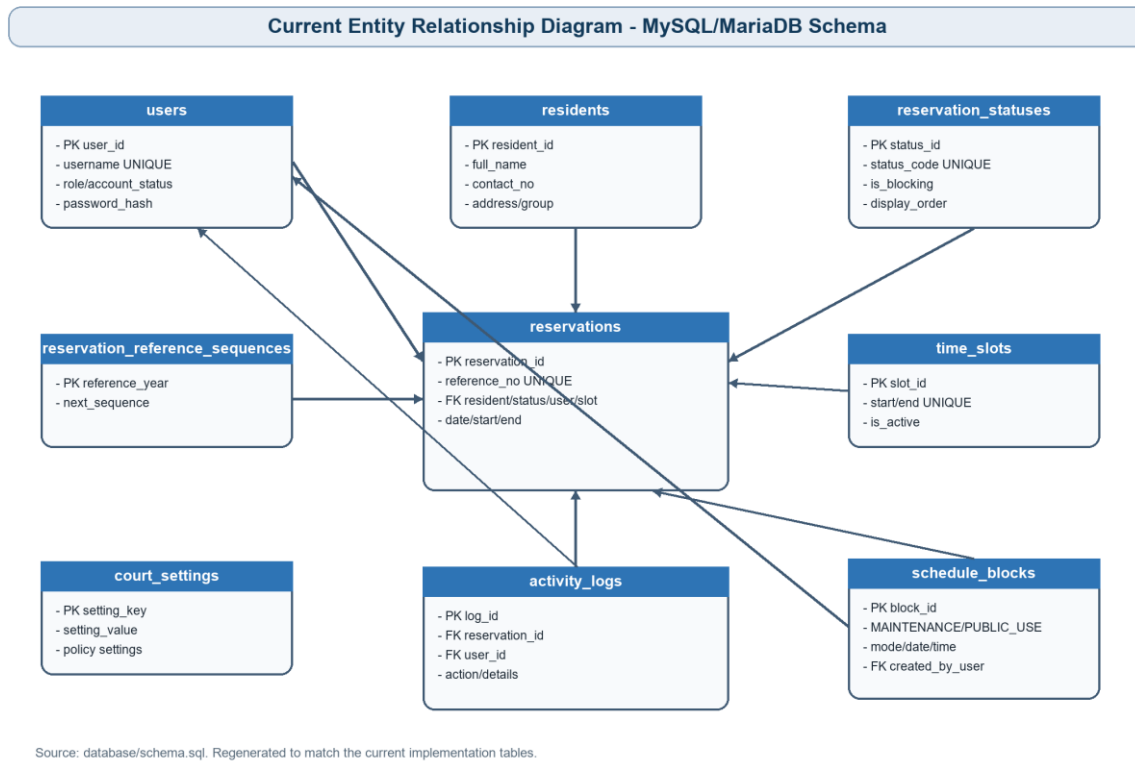


Figure 2.8. Entity Relationship Diagram of the Basketball Court Scheduling System

2.4 USER INTERFACE

Login Page

REPUBLICA NG PILIPINAS



Barangay Sto. Niño
CITY OF PARAÑAQUE

Welcome.
Magandang hapon.

Sign in to start the basketball court reservation system.
Mag-sign in upang magsimula.

TODAY
Martes · ika-19 ng Mayo, 2026 · 5:44 PM

● OFFICE COMPUTER · SYSTEM READY · V1.0

Sign in

Mag-sign in gamit ang iyong account.

Username · Pangalan ng user

Password · Password

Sign in

Forgot password?
Ask the administrator to reset your account at the barangay office.
Only barangay personnel can use this system.

Figure 2.9. Login Page

The login page is used by authorized Admin and Staff accounts before accessing the office system.

Dashboard

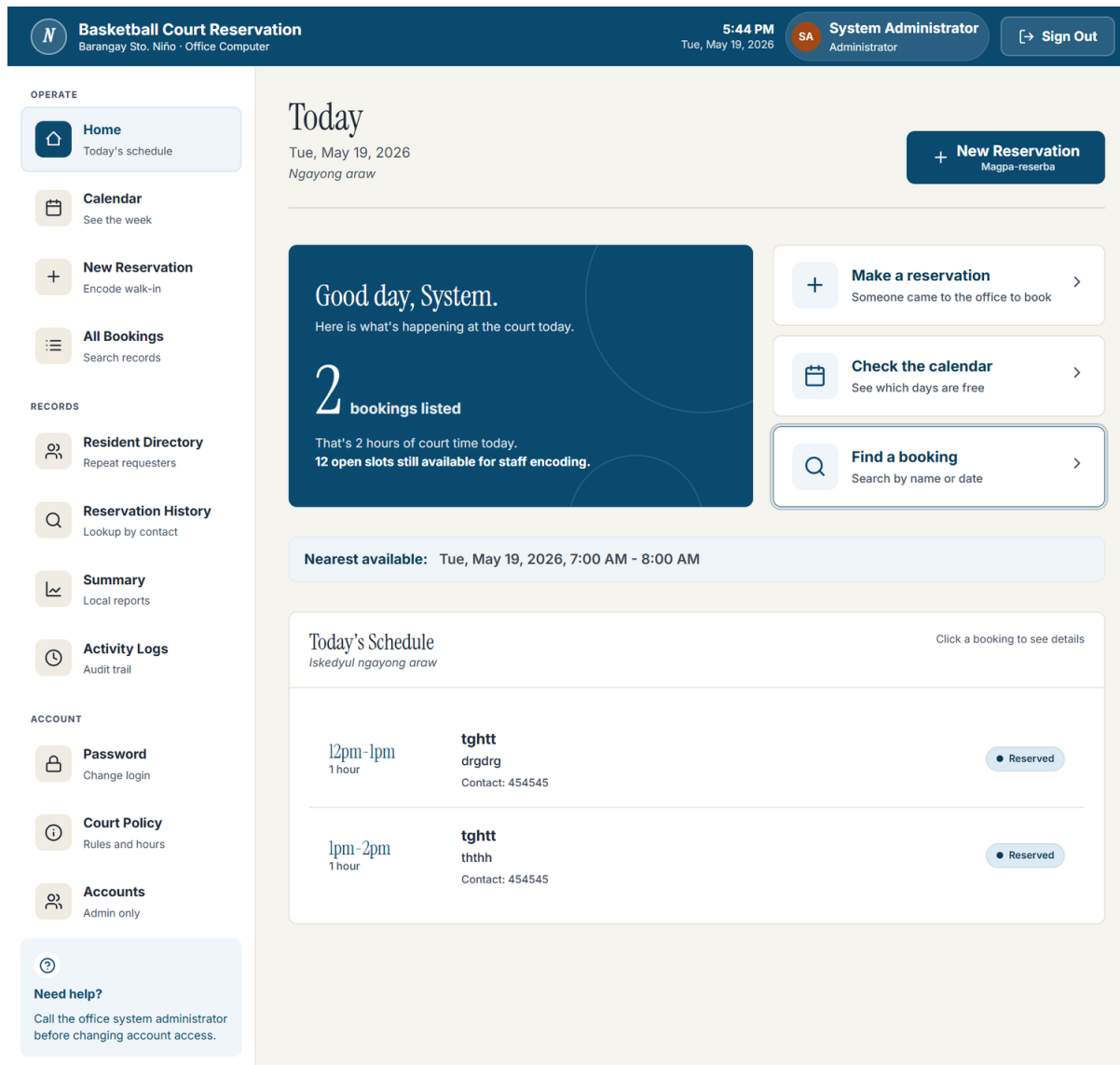


Figure 2.10. Dashboard

The dashboard summarizes today's schedule, alerts, counts, and nearest available slot information for quick office checking.

Calendar/Schedule View

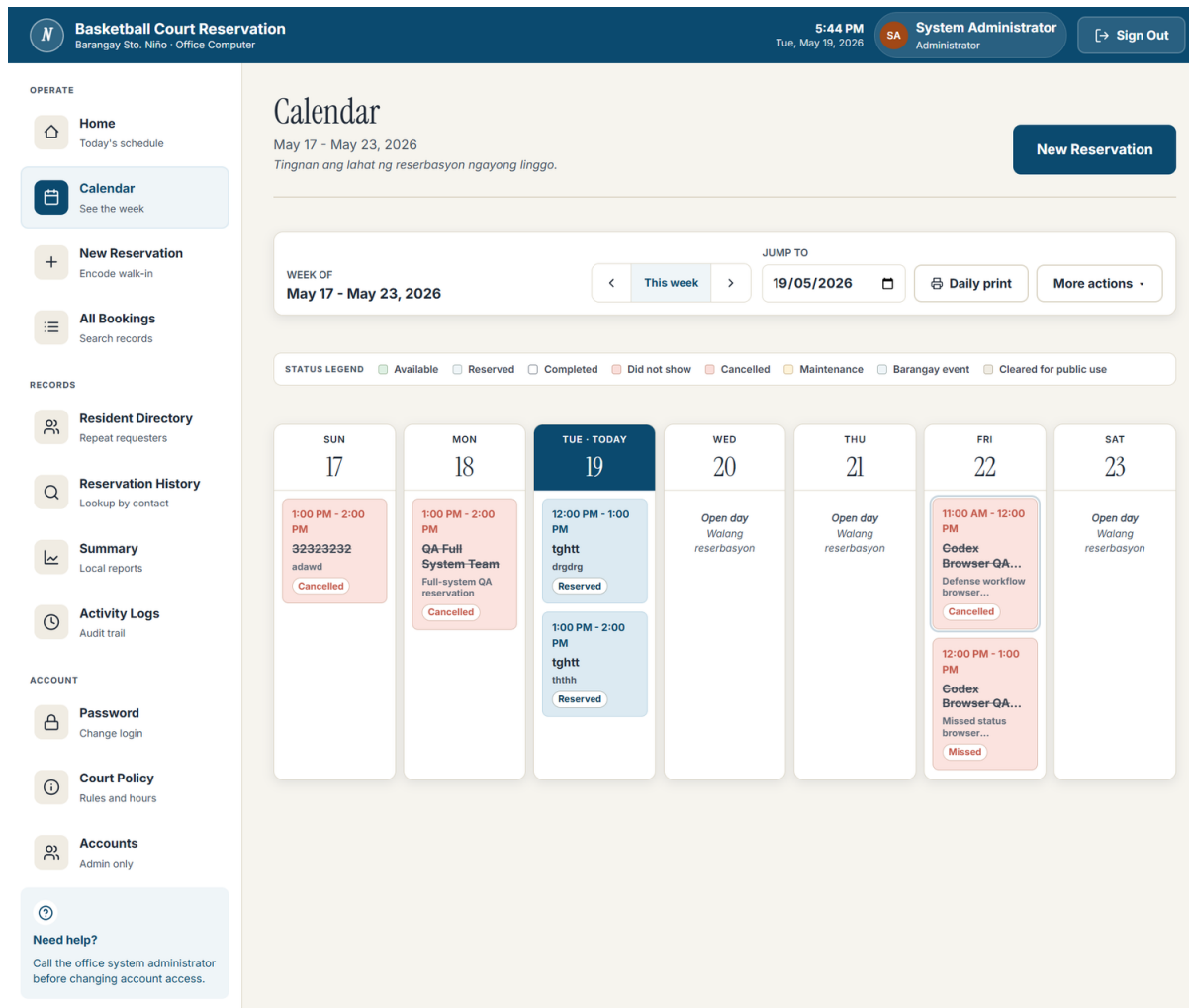


Figure 2.11. Calendar/Schedule View

The schedule view displays available, reserved, missed, cancelled, completed, maintenance, and public-use blocked slots.

New Reservation / Reservation Form

The screenshot displays the 'Basketball Court Reservation' web application. The top header includes the system name, location (Barangay Sto. Niño - Office Computer), time (5:45 PM), date (Tue, May 19, 2026), and user role (System Administrator). A sidebar on the left lists navigation options under 'OPERATE' (Home, Calendar, New Reservation, All Bookings) and 'RECORDS' (Resident Directory, Reservation History, Summary, Activity Logs). The main area shows a 'Calendar' for May 17-23, 2026, with a 'New Reservation' button. A modal window titled 'RESERVATION #66' is open, showing details for a reservation made by 'tghtt' at 12:00 PM - 1:00 PM. The form fields include Requester or group, Contact, Address, Purpose, and Remarks. At the bottom of the modal are buttons for 'Edit', 'Print slip', 'Mark as done', 'Mark as missed', and 'Cancel'. The background calendar shows a grid with dates and reservation status indicators like 'Reserved' and 'Missed'.

Basketball Court Reservation
Barangay Sto. Niño - Office Computer

5:45 PM
Tue, May 19, 2026

SA System Administrator
Administrator

[> Sign Out]

OPERATE

- Home
Today's schedule
- Calendar
See the week
- New Reservation
Encode walk-in
- All Bookings
Search records

RECORDS

- Resident Directory
Repeat requesters
- Reservation History
Lookup by contact
- Summary
Local reports
- Activity Logs
Audit trail

ACCOUNT

- Password
Change login
- Court Policy
Rules and hours
- Accounts
Admin only

Need help?
Call the office system administrator before changing account access.

Calendar
May 17 - May 23, 2026
Tingnan ang lahat ng reserbasyon ngayong linggo.

RESERVATION #66

tghtt
drgdrg

Time 12:00 PM - 1:00 PM

Requester or group tghtt

Contact 454545

Address htth

Purpose drgdrg

Remarks No remarks recorded.

Edit Print slip Mark as done Mark as missed Cancel

WEEK OF May 1

STATUS

Barangay event ☐ Cleared for public use

FRI 22 SAT 23

11:00 AM - 12:00 PM
Codex Browser QA...
Defense workflow browser...
Cancelled

12:00 PM - 1:00 PM
Codex Browser QA...
Missed status browser...
Missed

Open day
Walang reserbasyon

Figure 2.12. New Reservation / Reservation Form

The reservation form is used by barangay personnel to encode resident/requester details, date, time, and purpose.

Reservation Details

The screenshot displays the 'Basketball Court Reservation' web application. The top header shows the system name, location (Barangay Sto. Niño - Office Computer), time (5:45 PM), date (Tue, May 19, 2026), and user role (System Administrator). A sidebar on the left contains navigation links for Home, Calendar, New Reservation, All Bookings, Resident Directory, Reservation History, Summary, Activity Logs, Password, Court Policy, and Accounts. The main content area features a 'Calendar' view for May 17 - May 23, 2026, with a 'New Reservation' button. A modal window titled 'RESERVATION #66' is open, showing details for a reservation made by 'tghtt' on May 19, 2026, at 12:00 PM - 1:00 PM. The reservation is marked as 'Reserved'. The modal includes fields for Reference number (BCS-2026-000014), Date, Time, Requester or group, Contact, and Address, along with action buttons: Edit, Print slip, Mark as done, Mark as missed, and Cancel.

Basketball Court Reservation
Barangay Sto. Niño - Office Computer

5:45 PM
Tue, May 19, 2026

SA System Administrator
Administrator

[→] Sign Out

OPERATE

- Home
Today's schedule
- Calendar
See the week
- New Reservation
Encode walk-in
- All Bookings
Search records

RECORDS

- Resident Directory
Repeat requesters
- Reservation History
Lookup by contact
- Summary
Local reports
- Activity Logs
Audit trail

ACCOUNT

- Password
Change login
- Court Policy
Rules and hours
- Accounts
Admin only

Calendar
May 17 - May 23, 2026
Tingnan ang lahat ng reserbasyon ngayong linggo.

New Reservation

WEEK OF
May 17 - May 23, 2026

JUMP TO
19/05/2026

Daily print More actions

RESERVATION #66

tghtt
drgdrg

Reserved

Reference number BCS-2026-000014

Date Tue, May 19, 2026

Time 12:00 PM - 1:00 PM

Requester or group tghtt

Contact 454545

Address httht

Edit Print slip Mark as done Mark as missed Cancel

Barangay event Cleared for public use

FRI 22
11:00 AM - 12:00 PM
Codex Browser QA...
Defense workflow browser...
Cancelled

SAT 23
Open day
Walang reserbasyon

12:00 PM - 1:00 PM
Codex Browser QA...
Missed status browser...
Missed

Need help?
Call the office system administrator before changing account access.

Figure 2.13. Reservation Details

The detail drawer displays reservation reference information, requester data, schedule details, and status actions.

Reservation Records

Figure 2.14. Reservation Records

The records page supports searching, filtering, reviewing, printing, and exporting reservation data.

Account Management

Basketball Court Reservation
Barangay Sto. Niño · Office Computer

5:48 PM
Tue, May 19, 2026

SA System Administrator
Administrator

Sign Out

OPERATE

- Home
Today's schedule
- Calendar
See the week
- New Reservation
Encode walk-in
- All Bookings
Search records

RECORDS

- Resident Directory
Repeat requesters
- Reservation History
Lookup by contact
- Summary
Local reports
- Activity Logs
Audit trail

ACCOUNT

- Password
Change login
- Court Policy
Rules and hours
- Accounts**
Admin only
- Need help?
Call the office system administrator before changing account access.

ADMIN

Accounts

Create local logins and control who can use this office device.

1 active staff, 1 admin · 2 inactive

Signed in as System Administrator. The current account cannot be deactivated from this screen.

NEW ACCOUNT

Create local login

Use names staff recognize at the desk. Passwords stay local to this installation.

Full name · Buong pangalan

Username · Pangalan ng user

Password · Password

At least 8 characters, mixing letters and numbers.

Role · Tungkulin

Staff

Create Account

Local users

3 accounts saved on this system

FULL NAME	USERNAME	ROLE	STATUS	CREATED	ACTION
System Administrator	admin	Admin	Active	May 13, 2026, 3:44 AM	Current account
QA Staff 366571	qa_staff_366571	Staff	Inactive	May 16, 2026, 6:12 PM	Activate
QA Staff Role Check	qa_staff_357518	Staff	Inactive	May 18, 2026, 12:59 PM	Activate

Figure 2.15. Account Management

The account management screen is restricted to Admin users and supports Admin/Staff account handling.

Activity Logs

OPERATE

Home

Today's schedule

Calendar

See the week

New Reservation

Encode walk-in

All Bookings

Search records

RECORDS

Resident Directory

Repeat requesters

Reservation History

Lookup by contact

Summary

Local reports

Activity Logs

Audit trail

ACCOUNT

Password

Change login

AUDIT TRAIL

Activity logs

Search staff actions recorded by the local system.
Tala ng ginawa sa opisina.

Find a recorded action

Use this when a resident asks what changed, who changed it, or when it happened.

111 shown

All Today This week

Search name, action, or details

User, action, or details

Action

All actions

Date recorded

dd/mm/yyyy

Apply

Clear

Recorded actions

111 log rows from the local audit table

Export CSV

DATE AND TIME	USER	ACTION	DETAILS
May 19, 2026, 5:49 PM	Codex Staff Browser 1748	Login	User logged in.
May 19, 2026, 5:48 PM	System Administrator	Logout	User logged out.
May 19, 2026, 5:48 PM	System Administrator	Created an account	Created STAFF account for Codex Staff Browser 1748 (codex_staff_1748).
May 19, 2026, 5:47 PM	System Administrator	Cancelled the booking	Reservation status changed to CANCELLED.
May 19, 2026, 5:46 PM	System Administrator	Created a reservation	Created reservation BCS-2026-000015 for Codex Browser Verification 1746 on 2026-05-19 18:00-19:00.
May 19, 2026, 5:44 PM	System Administrator	Login	User logged in.
May 19, 2026, 5:43 PM	System Administrator	Login	User logged in.
May 19, 2026, 5:40 PM	System	Backup database	Created database backup barangay_court_scheduler_2026-05-19_174029.sql.
May 19, 2026, 5:39 PM	System Administrator	Login	User logged in.
May 19, 2026, 4:03 PM	System Administrator	Login	User logged in.
May 19, 2026, 12:08 PM	System Administrator	Login	User logged in.
May 19, 2026, 11:57 AM	System Administrator	Created a reservation	Created reservation BCS-2026-000014 for tghtt on 2026-05-19 12:00-13:00.
May 19, 2026, 11:56 AM	System Administrator	Created a reservation	Created reservation BCS-2026-000013 for tghtt on 2026-05-19 13:00-14:00.
May 19, 2026, 11:55 AM	System Administrator	Login	User logged in.
May 19, 2026, 11:44 AM	System Administrator	Login	User logged in.
May 19, 2026, 11:42 AM	System Administrator	Login	User logged in.
May 19, 2026, 11:35 AM	System Administrator	Login	User logged in.
May 19, 2026, 11:33 AM	System Administrator	Login	User logged in.
May 19, 2026, 11:27 AM	System Administrator	Login	User logged in.
May 19, 2026, 7:11 AM	System Administrator	Login	User logged in.
May 19, 2026, 6:09 AM	System Administrator	Login	User logged in.
May 19, 2026, 5:53 AM	System Administrator	Login	User logged in.

Figure 2.16. Activity Logs

The activity log screen shows the audit trail for reservation, account, court policy, and schedule-related actions.

Reports

Figure 2.17. Reports

The reports screen summarizes reservation counts, status breakdowns, usage, maintenance/public-use ranges, and exportable report data.

Maintenance Block Modal

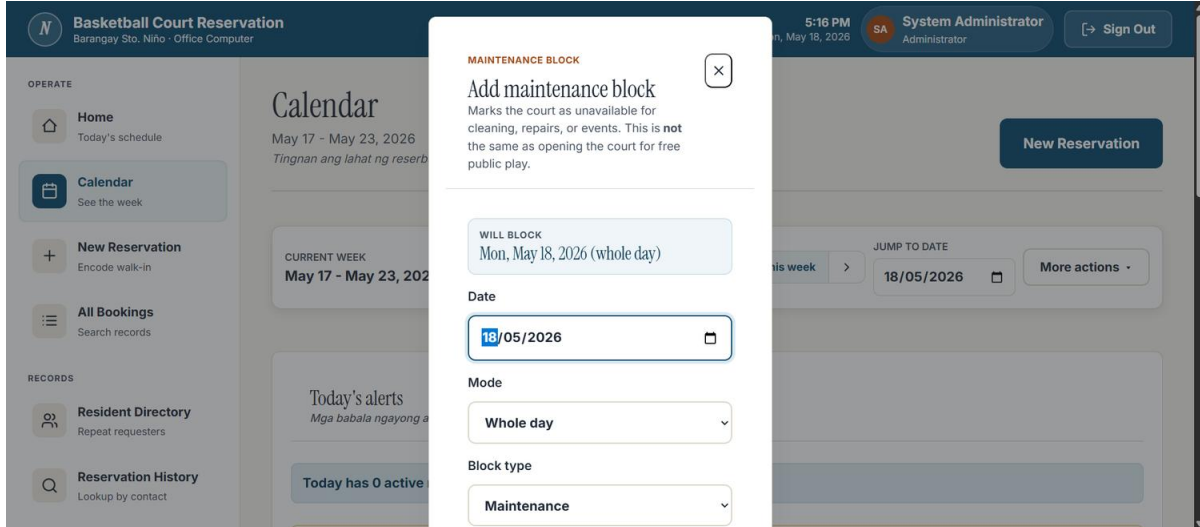


Figure 2.18. Maintenance Block Modal

The maintenance block modal marks unavailable court time; backup and restore are handled through separate Windows maintenance scripts.

Backup / Maintenance Tools

[INSERT SCREENSHOT: Maintenance Tools Launcher / Backup-Restore Tools]

Backup and restore are implemented as local Windows maintenance tools and scripts. No verified screenshot of the launcher was found among the source screenshots.

CHAPTER 3: THE FORMS

The following ASL forms are prepared in placeholder format because no completed signed forms were found in the source files.

3.1 ASL PLAN OF ACTIVITIES

Activity	Date	Objective	Participants	Remarks
Initial coordination with Barangay Sto. Niño	[INSERT DATE]	Confirm community need and project scope.	Group members; barangay personnel	[INSERT REMARKS]
Data gathering and process observation	[INSERT DATE]	Understand current court scheduling process and required details.	Group members; barangay personnel	[INSERT REMARKS]
System design and database planning	[INSERT DATE]	Prepare UI flow, database structure, and offline deployment plan.	Group members	[INSERT REMARKS]
Program development	[INSERT DATE]	Build login, reservation, schedule, account, reporting, and logging features.	Group members	[INSERT REMARKS]
Testing and verification	[INSERT DATE]	Verify overlap prevention, schedule display, UI screens, and offline readiness.	Group members	[INSERT REMARKS]
Presentation / defense preparation	[INSERT DATE]	Prepare ASL presentation, documentation, and evidence.	Group members	[INSERT REMARKS]
Turnover / user briefing	[INSERT DATE]	Brief barangay personnel and prepare acceptance records.	Group members; barangay personnel	[INSERT REMARKS]

3.2 COLLEGE DEAN REPORT

College	College of Science
School Year	[INSERT SCHOOL YEAR]
Semester	[INSERT SEMESTER]
Date	[INSERT DATE]
Project Title	Basketball Court Scheduling System
Community/Organization	Barangay Sto. Niño, Parañaque City
Report/Remarks	[INSERT VERIFIED COLLEGE DEAN REPORT DETAILS HERE]

3.3 FACULTY MEMBER REPORT FORM

Faculty Member	[INSERT FACULTY MEMBER NAME HERE]
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Subject/Section	[INSERT SUBJECT AND SECTION HERE]
Community/Organization	Barangay Sto. Niño, Parañaque City
Project Description	Offline barangay-office basketball court scheduling system.
Report/Remarks	[INSERT VERIFIED FACULTY MEMBER REPORT HERE]

3.4 FACULTY MEMBER EVALUATION FORM

Semester	[INSERT SEMESTER]
School Year	[INSERT SCHOOL YEAR]
Name of Professor	[INSERT PROFESSOR NAME HERE]
Name of Community/Organization Served	Barangay Sto. Niño, Parañaque City
Evaluation Scale	1 = Poor, 2 = Fair, 3 = Good, 4 = Very Good, 5 = Outstanding
Evaluator Name / Signature / Date	[INSERT VERIFIED EVALUATION DETAILS HERE]

3.5 STUDENT REGISTRATION FORM

Roviklein P. Fabre

Name of Student	Roviklein P. Fabre
Student No.	[INSERT STUDENT NUMBER HERE]
Course & Year	[INSERT COURSE AND YEAR LEVEL HERE]
Permanent Address	[INSERT ADDRESS HERE]
Telephone / CP No.	[INSERT CONTACT NUMBER HERE]
Subject with ASL	[INSERT SUBJECT HERE]
Section / Day / Time / Room	[INSERT CLASS DETAILS HERE]
Semester / School Year	[INSERT SEMESTER AND SCHOOL YEAR HERE]
Professor	[INSERT PROFESSOR NAME HERE]
Community / Organization	Barangay Sto. Niño, Parañaque City
Community Address / Contact Person	[INSERT VERIFIED BARANGAY CONTACT DETAILS HERE]

Jacob Ethan R. Gabaldon

Name of Student	Jacob Ethan R. Gabaldon
Student No.	[INSERT STUDENT NUMBER HERE]
Course & Year	[INSERT COURSE AND YEAR LEVEL HERE]
Permanent Address	[INSERT ADDRESS HERE]
Telephone / CP No.	[INSERT CONTACT NUMBER HERE]
Subject with ASL	[INSERT SUBJECT HERE]
Section / Day / Time / Room	[INSERT CLASS DETAILS HERE]
Semester / School Year	[INSERT SEMESTER AND SCHOOL YEAR HERE]
Professor	[INSERT PROFESSOR NAME HERE]
Community / Organization	Barangay Sto. Niño, Parañaque City
Community Address / Contact Person	[INSERT VERIFIED BARANGAY CONTACT DETAILS HERE]

Peter T. Guo

Name of Student	Peter T. Guo
Student No.	[INSERT STUDENT NUMBER HERE]
Course & Year	[INSERT COURSE AND YEAR LEVEL HERE]
Permanent Address	[INSERT ADDRESS HERE]
Telephone / CP No.	[INSERT CONTACT NUMBER HERE]
Subject with ASL	[INSERT SUBJECT HERE]
Section / Day / Time / Room	[INSERT CLASS DETAILS HERE]
Semester / School Year	[INSERT SEMESTER AND SCHOOL YEAR HERE]
Professor	[INSERT PROFESSOR NAME HERE]
Community / Organization	Barangay Sto. Niño, Parañaque City
Community Address / Contact Person	[INSERT VERIFIED BARANGAY CONTACT DETAILS HERE]

Zedrick Lonzaga

Name of Student	Zedrick Lonzaga
Student No.	[INSERT STUDENT NUMBER HERE]
Course & Year	[INSERT COURSE AND YEAR LEVEL HERE]
Permanent Address	[INSERT ADDRESS HERE]
Telephone / CP No.	[INSERT CONTACT NUMBER HERE]
Subject with ASL	[INSERT SUBJECT HERE]
Section / Day / Time / Room	[INSERT CLASS DETAILS HERE]
Semester / School Year	[INSERT SEMESTER AND SCHOOL YEAR HERE]
Professor	[INSERT PROFESSOR NAME HERE]
Community / Organization	Barangay Sto. Niño, Parañaque City
Community Address / Contact Person	[INSERT VERIFIED BARANGAY CONTACT DETAILS HERE]

Benedict Riley D. Quizon

Name of Student	Benedict Riley D. Quizon
Student No.	[INSERT STUDENT NUMBER HERE]
Course & Year	[INSERT COURSE AND YEAR LEVEL HERE]
Permanent Address	[INSERT ADDRESS HERE]
Telephone / CP No.	[INSERT CONTACT NUMBER HERE]
Subject with ASL	[INSERT SUBJECT HERE]
Section / Day / Time / Room	[INSERT CLASS DETAILS HERE]
Semester / School Year	[INSERT SEMESTER AND SCHOOL YEAR HERE]
Professor	[INSERT PROFESSOR NAME HERE]
Community / Organization	Barangay Sto. Niño, Parañaque City
Community Address / Contact Person	[INSERT VERIFIED BARANGAY CONTACT DETAILS HERE]

Rhegille Gabriel L. Rodriguez

Name of Student	Rhegille Gabriel L. Rodriguez
Student No.	[INSERT STUDENT NUMBER HERE]
Course & Year	[INSERT COURSE AND YEAR LEVEL HERE]
Permanent Address	[INSERT ADDRESS HERE]
Telephone / CP No.	[INSERT CONTACT NUMBER HERE]
Subject with ASL	[INSERT SUBJECT HERE]
Section / Day / Time / Room	[INSERT CLASS DETAILS HERE]
Semester / School Year	[INSERT SEMESTER AND SCHOOL YEAR HERE]
Professor	[INSERT PROFESSOR NAME HERE]
Community / Organization	Barangay Sto. Niño, Parañaque City
Community Address / Contact Person	[INSERT VERIFIED BARANGAY CONTACT DETAILS HERE]

3.6 CLASS EVALUATION FORM

Subject	[INSERT SUBJECT HERE]
Name of Professor	[INSERT PROFESSOR NAME HERE]
Department	Computer Science
College	College of Science
Community/Organization Served	Barangay Sto. Niño, Parañaque City
Class Evaluation	[INSERT VERIFIED CLASS EVALUATION FORM HERE]

3.7 STUDENT EVALUATION FORM

Roviklein P. Fabre

Name of Student	Roviklein P. Fabre
Student No.	[INSERT STUDENT NUMBER HERE]
Course / Year Level	[INSERT COURSE AND YEAR LEVEL HERE]
Subject	[INSERT SUBJECT HERE]
Professor	[INSERT PROFESSOR NAME HERE]
Community / Organization Served	Barangay Sto. Niño, Parañaque City
Scale	1 = Strongly Disagree, 2 = Disagree, 3 = Neither Agree nor Disagree, 4 = Agree, 5 = Strongly Agree
Preparation and Support	1 2 3 4 5 [INSERT RESPONSES]

Cognitive Learning	1 2 3 4 5 [INSERT RESPONSES]
Affective Learning	1 2 3 4 5 [INSERT RESPONSES]
Open-Ended Reflection	[INSERT VERIFIED STUDENT RESPONSE HERE]

Jacob Ethan R. Gabaldon

Name of Student	Jacob Ethan R. Gabaldon
Student No.	[INSERT STUDENT NUMBER HERE]
Course / Year Level	[INSERT COURSE AND YEAR LEVEL HERE]
Subject	[INSERT SUBJECT HERE]
Professor	[INSERT PROFESSOR NAME HERE]
Community / Organization Served	Barangay Sto. Niño, Parañaque City
Scale	1 = Strongly Disagree, 2 = Disagree, 3 = Neither Agree nor Disagree, 4 = Agree, 5 = Strongly Agree
Preparation and Support	1 2 3 4 5 [INSERT RESPONSES]
Cognitive Learning	1 2 3 4 5 [INSERT RESPONSES]
Affective Learning	1 2 3 4 5 [INSERT RESPONSES]
Open-Ended Reflection	[INSERT VERIFIED STUDENT RESPONSE HERE]

Peter T. Guo

Name of Student	Peter T. Guo
Student No.	[INSERT STUDENT NUMBER HERE]
Course / Year Level	[INSERT COURSE AND YEAR LEVEL HERE]
Subject	[INSERT SUBJECT HERE]
Professor	[INSERT PROFESSOR NAME HERE]
Community / Organization Served	Barangay Sto. Niño, Parañaque City
Scale	1 = Strongly Disagree, 2 = Disagree, 3 = Neither Agree nor Disagree, 4 = Agree, 5 = Strongly Agree
Preparation and Support	1 2 3 4 5 [INSERT RESPONSES]
Cognitive Learning	1 2 3 4 5 [INSERT RESPONSES]
Affective Learning	1 2 3 4 5 [INSERT RESPONSES]
Open-Ended Reflection	[INSERT VERIFIED STUDENT RESPONSE HERE]

Zedrick Lonzaga

Name of Student	Zedrick Lonzaga
Student No.	[INSERT STUDENT NUMBER HERE]
Course / Year Level	[INSERT COURSE AND YEAR LEVEL HERE]
Subject	[INSERT SUBJECT HERE]
Professor	[INSERT PROFESSOR NAME HERE]
Community / Organization Served	Barangay Sto. Niño, Parañaque City
Scale	1 = Strongly Disagree, 2 = Disagree, 3 = Neither Agree nor Disagree, 4 = Agree, 5 = Strongly Agree
Preparation and Support	1 2 3 4 5 [INSERT RESPONSES]
Cognitive Learning	1 2 3 4 5 [INSERT RESPONSES]
Affective Learning	1 2 3 4 5 [INSERT RESPONSES]
Open-Ended Reflection	[INSERT VERIFIED STUDENT RESPONSE HERE]

Benedict Riley D. Quizon

Name of Student	Benedict Riley D. Quizon
Student No.	[INSERT STUDENT NUMBER HERE]
Course / Year Level	[INSERT COURSE AND YEAR LEVEL HERE]
Subject	[INSERT SUBJECT HERE]
Professor	[INSERT PROFESSOR NAME HERE]
Community / Organization Served	Barangay Sto. Niño, Parañaque City
Scale	1 = Strongly Disagree, 2 = Disagree, 3 = Neither Agree nor Disagree, 4 = Agree, 5 = Strongly Agree
Preparation and Support	1 2 3 4 5 [INSERT RESPONSES]
Cognitive Learning	1 2 3 4 5 [INSERT RESPONSES]
Affective Learning	1 2 3 4 5 [INSERT RESPONSES]
Open-Ended Reflection	[INSERT VERIFIED STUDENT RESPONSE HERE]

Rhegille Gabriel L. Rodriguez

Name of Student	Rhegille Gabriel L. Rodriguez
Student No.	[INSERT STUDENT NUMBER HERE]
Course / Year Level	[INSERT COURSE AND YEAR LEVEL HERE]
Subject	[INSERT SUBJECT HERE]
Professor	[INSERT PROFESSOR NAME HERE]
Community / Organization Served	Barangay Sto. Niño, Parañaque City
Scale	1 = Strongly Disagree, 2 = Disagree, 3 = Neither Agree nor Disagree, 4 = Agree, 5 = Strongly Agree
Preparation and Support	1 2 3 4 5 [INSERT RESPONSES]
Cognitive Learning	1 2 3 4 5 [INSERT RESPONSES]
Affective Learning	1 2 3 4 5 [INSERT RESPONSES]
Open-Ended Reflection	[INSERT VERIFIED STUDENT RESPONSE HERE]

3.8 COMMUNITY/ORGANIZATION EVALUATION FORM

Community / Organization	Barangay Sto. Niño, Parañaque City
Project Title	Basketball Court Scheduling System
Evaluator Name	[INSERT VERIFIED BARANGAY EVALUATOR NAME HERE]
Position	[INSERT VERIFIED POSITION HERE]
Date	[INSERT DATE HERE]
Evaluation / Comments	[INSERT VERIFIED COMMUNITY/ORGANIZATION EVALUATION HERE]
Signature	[INSERT SIGNATURE HERE]

3.9 TURNOVER / ACCEPTANCE FORM

Project Turned Over	Basketball Court Scheduling System
Receiving Community	Barangay Sto. Niño, Parañaque City
Items Turned Over	[INSERT SYSTEM FILES, INSTALLATION MEDIA, USER GUIDE, AND TRAINING MATERIALS HERE]
Receiving Representative	[INSERT VERIFIED NAME AND POSITION HERE]
Date of Turnover	[INSERT DATE HERE]
Remarks	[INSERT TURNOVER REMARKS HERE]
Signatures	[INSERT SIGNATURES HERE]

CHAPTER 4: REFLECTION

Personal reflections must be written or approved by the named students. The placeholders below provide prompts only and should not be treated as completed student reflections.

Roviklein P. Fabre

[INSERT PERSONAL ASL REFLECTION HERE]

Suggested reflection points:

- Experience working with Barangay Sto. Niño.
- What was learned from developing the Basketball Court Scheduling System.
- Challenges during planning, development, documentation, testing, or deployment.
- Technical skills developed in database design, backend/frontend development, verification, or documentation.
- Value of community service and applying computer science to real barangay needs.
- Recommendations for future improvement of the system or ASL process.

[INSERT PERSONAL ASL REFLECTION HERE]

Suggested reflection points:

- Experience working with Barangay Sto. Niño.
- What was learned from developing the Basketball Court Scheduling System.
- Challenges during planning, development, documentation, testing, or deployment.
- Technical skills developed in database design, backend/frontend development, verification, or documentation.
- Value of community service and applying computer science to real barangay needs.
- Recommendations for future improvement of the system or ASL process.

[INSERT PERSONAL ASL REFLECTION HERE]

Suggested reflection points:

- Experience working with Barangay Sto. Niño.
- What was learned from developing the Basketball Court Scheduling System.
- Challenges during planning, development, documentation, testing, or deployment.
- Technical skills developed in database design, backend/frontend development, verification, or documentation.
- Value of community service and applying computer science to real barangay needs.
- Recommendations for future improvement of the system or ASL process.

[INSERT PERSONAL ASL REFLECTION HERE]

Suggested reflection points:

- Experience working with Barangay Sto. Niño.
- What was learned from developing the Basketball Court Scheduling System.
- Challenges during planning, development, documentation, testing, or deployment.
- Technical skills developed in database design, backend/frontend development, verification, or documentation.
- Value of community service and applying computer science to real barangay needs.
- Recommendations for future improvement of the system or ASL process.

[INSERT PERSONAL ASL REFLECTION HERE]

Suggested reflection points:

- Experience working with Barangay Sto. Niño.
- What was learned from developing the Basketball Court Scheduling System.
- Challenges during planning, development, documentation, testing, or deployment.
- Technical skills developed in database design, backend/frontend development, verification, or documentation.
- Value of community service and applying computer science to real barangay needs.
- Recommendations for future improvement of the system or ASL process.

[INSERT PERSONAL ASL REFLECTION HERE]

Suggested reflection points:

- Experience working with Barangay Sto. Niño.
- What was learned from developing the Basketball Court Scheduling System.
- Challenges during planning, development, documentation, testing, or deployment.
- Technical skills developed in database design, backend/frontend development, verification, or documentation.
- Value of community service and applying computer science to real barangay needs.
- Recommendations for future improvement of the system or ASL process.

CHAPTER 5: DOCUMENTATION

This chapter is reserved for actual ASL photo documentation. No verified barangay visit, signing, development, testing, deployment, turnover, or training photos were found in the inspected source folders, so clearly labeled placeholders are provided.

5.1 Barangay Sto. Niño Barangay Hall

[INSERT PHOTO: Barangay Hall]

Barangay Sto. Niño barangay hall or official project location.

Barangay Sto. Niño barangay hall or official project location.

5.2 First Community Visit / Initial Coordination

[INSERT PHOTO: First Community Visit]

Initial coordination with barangay personnel regarding the court scheduling need.

Initial coordination with barangay personnel regarding the court scheduling need.

5.3 Data Gathering / Observation of Current Process

[INSERT PHOTO: Data Gathering / Current Process Observation]

Observation of current reservation and schedule-checking process.

Observation of current reservation and schedule-checking process.

5.4 Contract Signing / Approval / Coordination

[INSERT PHOTO: Contract Signing / Approval / Coordination]

Project approval, coordination, or ASL agreement documentation.

Project approval, coordination, or ASL agreement documentation.

5.5 Program Development

[INSERT PHOTO: Program Development]

Development of the offline Basketball Court Scheduling System.

5.6 System Testing

[INSERT PHOTO: System Testing]

Testing of reservation creation, schedule monitoring, overlap prevention, accounts, and activity logs.

Testing of reservation creation, schedule monitoring, overlap prevention, accounts, and activity logs.

5.7 ASL Presentation / Defense Preparation

[INSERT PHOTO: ASL Presentation / Defense Preparation]

Preparation of system presentation and ASL documentation.

Preparation of system presentation and ASL documentation.

5.8 Deployment / Turnover

[INSERT PHOTO: Deployment / Turnover]

Installation or turnover of the system to Barangay Sto. Niño personnel.

Installation or turnover of the system to Barangay Sto. Niño personnel.

5.9 User Training / Barangay Personnel Briefing

[INSERT PHOTO: User Training / Barangay Personnel Briefing]

Briefing authorized barangay personnel on how to use the installed system.

Briefing authorized barangay personnel on how to use the installed system.